E-Skin Report Spring 2026

By Cole Abbott

Summary

This spring I continued working on the E-Skin project under Shihan Lu. Last quarter we left off with a working data collection pipeline, and this quarter involved collecting data from the skin, setting up a 3D printer based testbed, and debugging the new PCB design. The initial data was not as good as we had hoped, and we were not able to extract the force and location from the data from the transducers. We tested different transducers, and evaluated their performance. Unfortunately we did not get a demo working, as we were not able to reliably detect sound wave reflections, but we made good progress with the hardware working reliably, and data collected across multiple transducer arrays.

Transducer Testing

Once the data collection pipeline was working well, we were able to collect data from the transducers. However, the data had more noise and the sound waves reflected off of a depression in the skin were not apparent.

Ideally, when there is nothing touching the skin, the data from the transducer should look like a decaying sine wave, with a rise in amplitude near t = 80us, which represents the sound waves reflecting off the end of the skin. However in reality, the reflection off the end of the skin often does not appear, and there are other peaks at varying points, which is not consistent across different transducers.

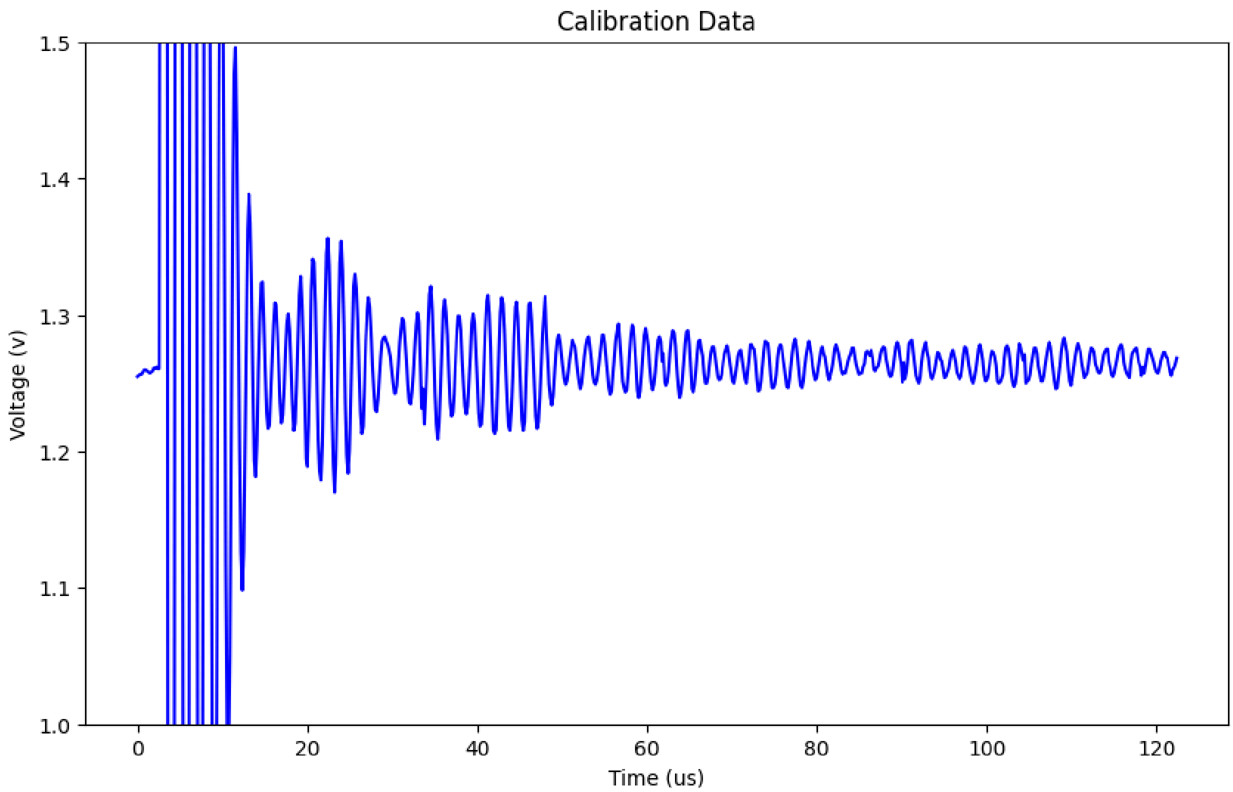

Figure 1. Data from a transducer with no pressure on skin

In figure 1, we see high amplitude ringing after the transducer is excited by the voltage pulse, which is expected, and dies down after ~10 us. Then we see a rise in amplitude just after 20us, which should correspond to ~1.5cm into the skin, however, there is no pressure applied there and nothing that the soundwave should reflect off of. It is unclear what is causing this peak, it is possible that it is due to sound reflecting around in the transducer array, or that it is caused by the interface between the array and the skin.

Additionally, in this plot we do not see any sound reflected off the end of the skin. It is possible that this is because the signal is attenuated too much and can no longer be detected.

After seeing that the first transducer array did not perform as well as we hoped, Shihan fabricated 2 more arrays with different transducers, a small transducer and a large transducer. These transducers were able to detect the edge of the skin, and detect presses with some success.

Small Transducer Array

The small transducer was tested with a smaller 32mm skin section. The speed of sound in SimuGel #0 is 1449 m/s, so the theoretical time for a sound wave to travel to the edge and back is 44.17 us.

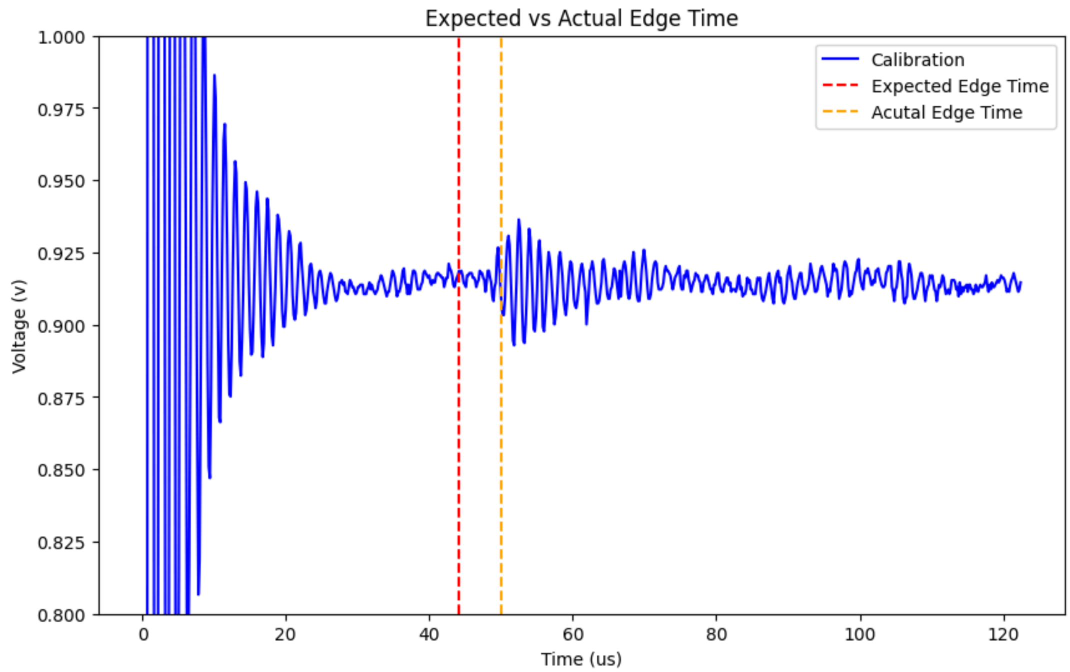

Figure 2 Small transducer edge reflection

In Figure 2, we see a peak start at around 50us, this discrepancy can be explained by the fact that at t = 0, the pulse is delivered to the transducer, but the sound wave has not been emitted yet, and it takes some time for the transducer to start oscillating and emit the sound wave.

The small transducer array is able to detect the reflection off the edge of the skin with reasonable accuracy. However the amplitude of the reflected signal is very small, only ~0.05v when the signal reaches the PIC, which is after a 50x amplification from the receiver circuit. This makes the signal more sensitive to noise, and harder to detect the edge for large skin sizes.

Since we are able to detect a signal reflected off the edge of the skin, we now can test if we can detect a signal reflected off a probe pressed into the skin.

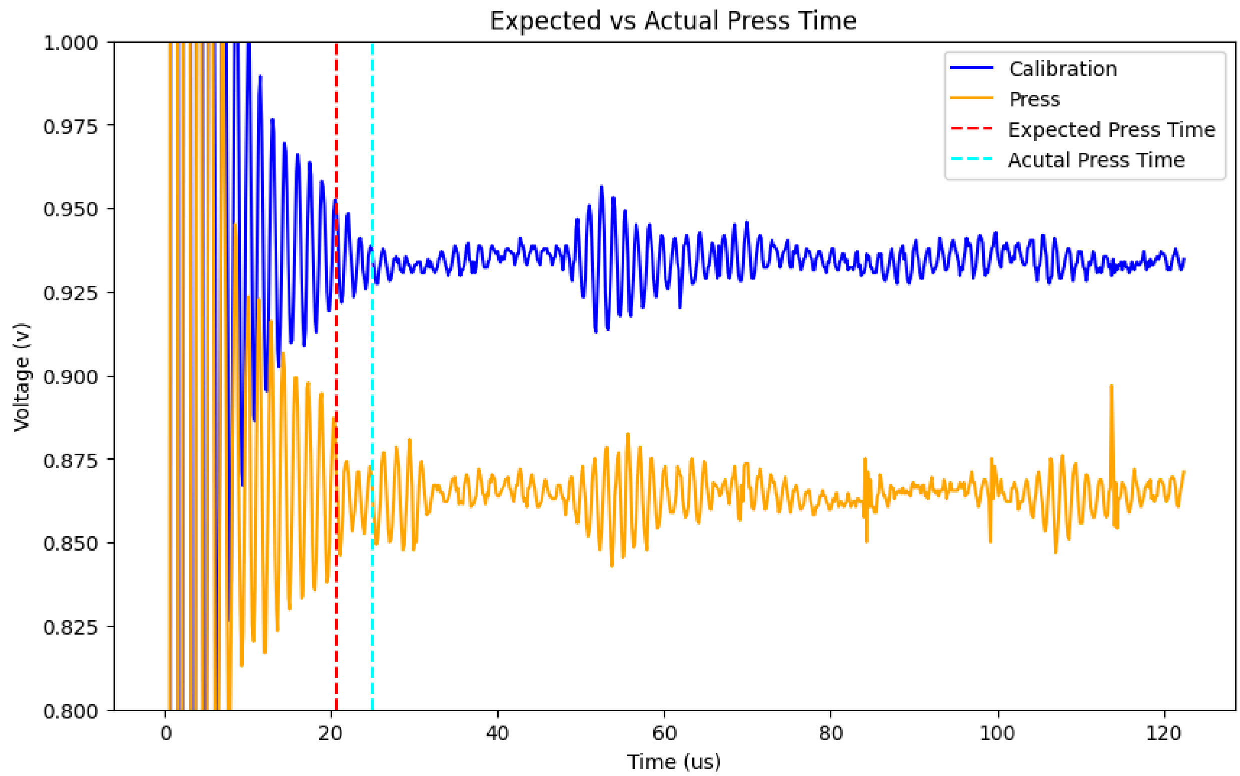

Figure 3 Small transducer with force applied

In Figure 3, the data with no force applied is shown alongside the data from when a probe was pressed into the skin 15 mm from the transducers array. The plots are shifted apart for clarity.

We can see a smaller peak starting around 25us, which again is 5us later than the predicted time, which aligns with the previous theory about the sound wave delay.

This validates that the skin is able to detect the location of force applied to it in some capacity, however it is very sensitive to disturbances, and often the applied force is not clearly represented in the data.

Large Transducer Array

The large transducer has a much longer ringing period, but is also a more sensitive receiver. Due to the longer ringing period, the larger transducers do not work with the small skin section, and were tested on a larger skin section that is 70mm long. The expected edge time is 96us.

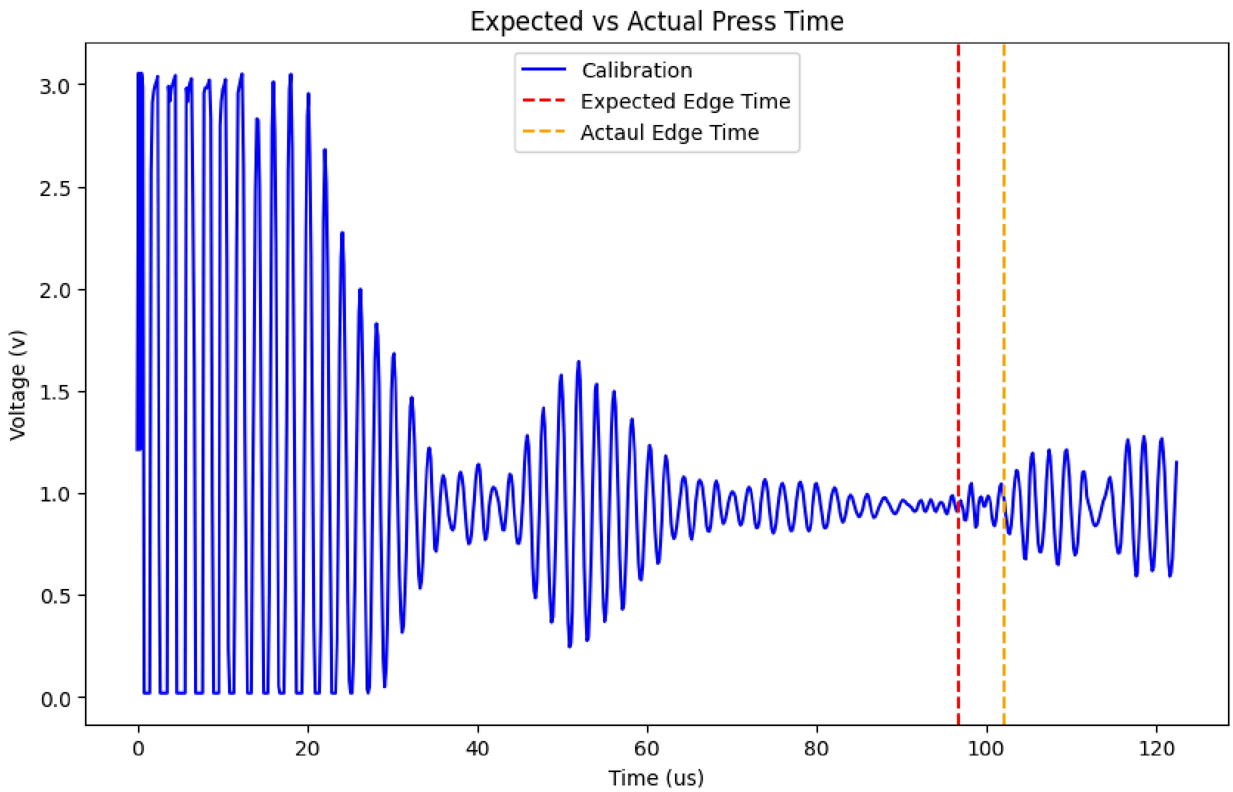

Figure 4 Large transducer array

In Figure 4, we can see that the large transducer has a ringing period of around 30us, which corresponds to a ~25mm deadzone where the transducer will not be able to detect any reflections. We also see a large peak at 50us, which again has no clear explanation. This peak does not change when pressing on the skin, but does change when adjusting the transducer skin interface, so it is likely caused by that somehow. At t = 100, we see the reflected signal from the edge of the skin, which again is delayed by 5us. Here the amplitude is much greater than the smaller transducers, with an amplitude of 0.7v, 14 times greater than the smaller transducer at double the range.

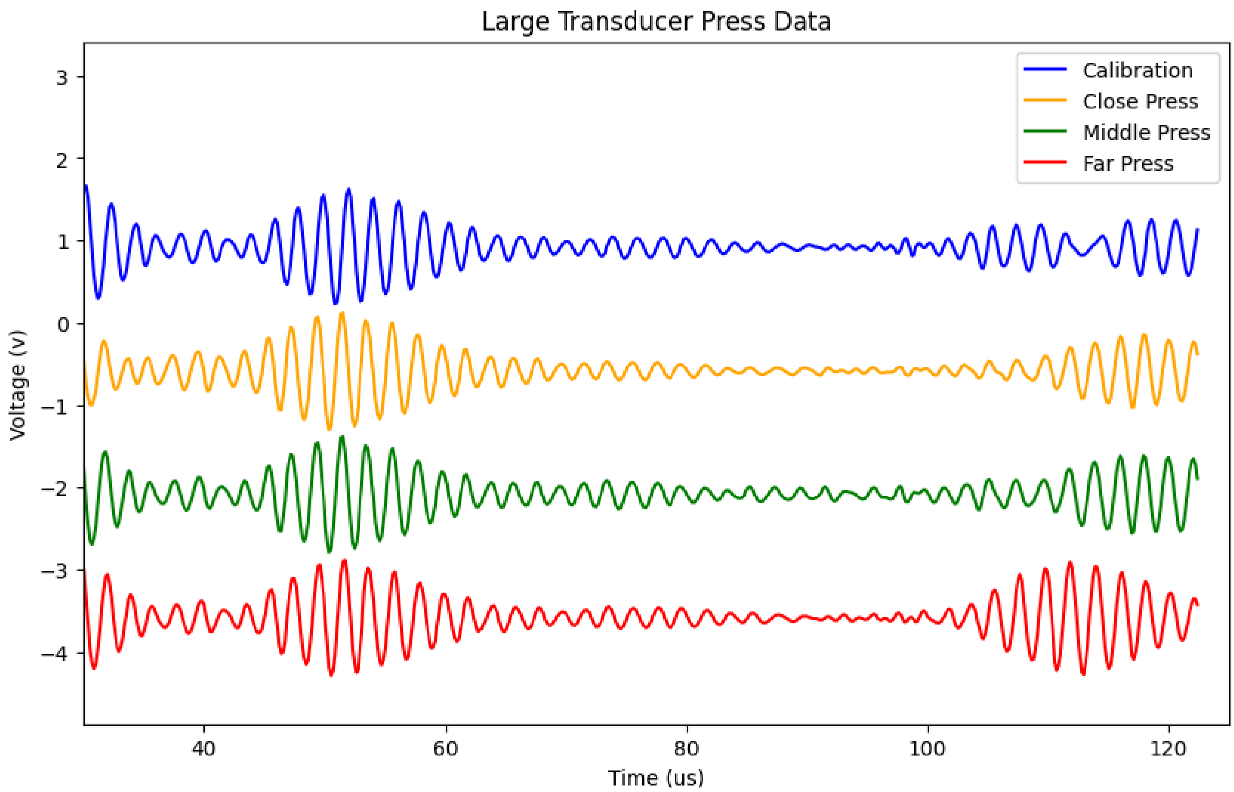

Figure 5 Large transducer with force applied

Figure 5 shows the large transducers with force applied at 3 points, close to the transducer, in the middle of the skin, and near the far edge. The ringing section is cropped for clarity, and the plots are shifted apart. Instead of seeing clear peaks at different time points, most of the signal does not change, except for the reflected edge, which is affected even when the force is not applied near the edge.

This implies that pressing on the skin can have an impact on the wave that reflects off the edge of the skin, rather than the desired behavior of reflecting a wave directly back.

New PCB

Shihan designed a new PBC layout to fix some of the issues with the first PCB, most importantly the PIC crashing, but also adding a synchronization signal between the Arduino and the PIC, and separating the control signals to the PIC, HV pulser, and multiplexers. The fix for the PIC crashing involved rerouting traces to avoid high noise areas, and adding a resistor in series with the signal to limit the current. See my report from winter quarter for more details on the PIC crashing issue.

Unfortunately, the rerouted traces did not stop the PIC from crashing when a 0 Ohm resistor was placed in series with the analog signal. I believe that the new PCB did successfully reduce the electrical interference between the pulser circuit and the analog signal, but due to how the receiver circuit is designed, it is still able to output a negative voltage. The bias tee is meant to shift the voltage into the 0-3.3v range, but in reality it is not able to shift the voltage high enough. I think that the best way to improve this design in the future is to use an op-amp that outputs 0-3.3v, instead of an op-amp that outputs -3.7-3.7v. For the time being, a resistor can be placed in series with the signal to limit the current, which prevents the PIC from crashing. The resistance is small enough that it does not have a significant effect on the quality of the signal.

Another feature of the new PCB is a synchronization signal between the Arduino and the PIC. Since the Arduino is responsible for setting the multiplexer channel, the PIC does not know what transducer the incoming data is coming from. In the past we had considered using an empty channel to detect when the cycle resets, but this is over complicated and not robust because if a connection to a transducer breaks, it will look like there are 2 empty channels, and we will get out of sync. The Arduino and PIC can be synchronized by adding another signal that will pulse once per full cycle. This will be timed so that the sync signal is high during the trigger interrupt, and the PIC can read the signal, and reset the channel counter when needed. The PIC can then add the channel count to the USB data so the PC knows which channel incoming data is from.

Test Setup

To calibrate and validate the skin, we integrated our skin into an existing tactile sensor testbed that uses a 3D printer to move a probe with a load cell to apply a force to the skin at different points. To integrate the eskin into this testing setup, we simply had to add our data collection python code to the existing code framework. The code for this can be found in the eskin_tactile_sensor branch of this tactile_testbed Github Repo.

The data from the transducers is very sensitive to the exact setup of the transducer skin interface, so in order to have consistent readings across test setups, we made a stand that holds the transducer array, and has adjustable screws to change the pressure on the skin. However, the clamping mechanism damaged the transducer array after leaving it installed for multiple days, so this should be redesigned in the future for better reliability.

Conclusions

Overall, I think the hardware, PIC and Arduino firmware, and data collection software are in a good place, and work reliably to give us the data we need. I think selecting a suitable transducer and fabricating a more robust transducer array will allow the skin to work in some capacity. I also think that using separate transducers for emitting the pulse and receiving the reflection could be beneficial. I think one of the reasons the large transducer is not able to detect reflections is because the ringing causes the soundwave to not be as sharp as it needs to be. Using a small transducer to emit a sharp pulse and a large transducer to record the response could be an effective approach, and I think this could be prototyped using 2 PCB boards without designing new hardware.